Milestone 1

Project Description

The goal of my project is to predict whether a person is likely to have diabetes based on a variety of health, lifestyle, and demographic factors. Instead of focusing on one single cause, I want to see how behaviors such as smoking, physical activity, diet, and access to care all combine to influence diabetes risk. My main research question is simple: *Can machine learning accurately identify people who are at higher risk for diabetes using everyday health indicators?*

The data I'm using comes from the **Behavioral Risk Factor Surveillance System (BRFSS 2015)**, published on Kaggle as the *Diabetes Health Indicators Dataset* by Alex Teboul. It contains more than **250,000 survey responses** from adults across the United States, each with **22 features** that describe things like BMI, age, cholesterol level, blood pressure, and whether the person exercises, smokes, or drinks alcohol. The column I'm trying to predict is called **Diabetes binary**, which is coded as **1 for "has diabetes"** and **0 for "does not."** Because the dataset is large and covers such a wide range of variables, it's perfect for comparing different models and learning which factors truly make the biggest difference.

The main groups who could benefit from this analysis are **public health officials, clinicians, and community organizations**. For public health agencies, a model like this could help identify which regions or populations might need more education or screening programs. Doctors and nurses could use similar approaches to flag high-risk patients earlier, while researchers might use the findings to study how social and lifestyle factors influence chronic disease.

This kind of work matters because diabetes is one of the most common and preventable, chronic conditions in the U.S. If we can better predict who's at risk using basic health information, it

could lead to earlier interventions, fewer hospitalizations, and lower costs for the healthcare system. Beyond science, the broader goal is social: helping people understand their own health risks and giving communities the tools to promote prevention rather than reaction.

Conclusions: Depth

I evaluated multiple models on the *diabetes.csv* dataset, with a binary target (any diabetes/prediabetes vs none). Two families stood out for how they control complexity and overfitting: **regularized logistic regression** (Week 2) and **tree-based ensembles** (Week 6).

Week 2 – Regularization (L1/L2/Elastic Net).

Logistic regression with **L2** and **L1** penalties both achieved **AUC = 0.818**, the best overall in my runs. These models reduce overfitting by **penalizing large coefficients**, which prevents the model from chasing noise, especially important here because several predictors are correlated (e.g., BMI, Age, HighBP). I also used **feature scaling** and **class_weight="balanced"** to stabilize optimization and handle class imbalance. Elastic Net (trained on a small feature subset for speed) reached **0.776**, illustrating that too-aggressive simplification (or too few features) can **underfit**. Compared to the scaled logistic baseline (**0.818**), the message is: **moderate regularization matches the best performance while keeping the model simple and interpretable**.

Week 6 – Trees & Random Forests.

A **Decision Tree** with shallow depth (max_depth=5, min_samples_leaf=10) reached **AUC = 0.796**. This depth/leaf control mitigates the tree's tendency to **overfit** tiny patterns. The **Random Forest** (bagging + feature subsampling) improved to **AUC = 0.818**, matching the best logistic models. Bagging reduces variance by averaging many diverse trees; the result is **strong**

generalization without heavy tuning. In practice, forests gave useful **feature importance** rankings (typically elevating BMI, Age, and blood-pressure variables), which aligned with intuition from EDA.

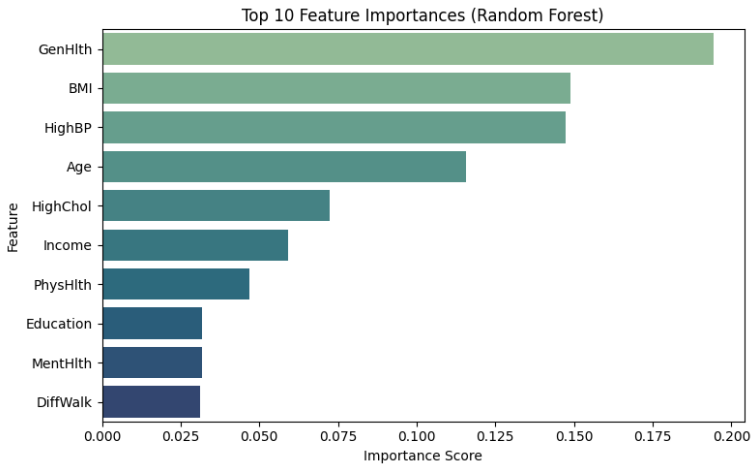

Figure 1. Top 10 features ranked by importance in Random Forest.

Evidence & synthesis.

- Best AUCs: **Logistic L1/L2 (0.818)** and **Random Forest (0.818)**.
- Overfitting controls that worked: **coefficient penalties (L1/L2)**, **balanced classes**, **scaling** for linear models; and **shallow depth / min leaf** plus **bagging** for trees.
- Underperformers showed why controls matter: **poly interactions (0.775)** and **tiny feature subsets (0.771–0.776)** leaned toward underfitting.

Conclusions: Breadth

Over the six-week modeling process, I applied and compared multiple machine-learning methods to the *diabetes.csv* dataset. Each week introduced a new concept for improving model accuracy,

interpretability, and generalization while addressing key issues like feature correlation and overfitting.

Week 1 – Polynomial & Interaction Terms

Adding squared and interaction features slightly improved the baseline logistic regression, reaching an AUC of ≈ 0.775. This showed that nonlinear relationships (e.g., between BMI and Age) exist, but adding too many terms can overfit noise if not paired with regularization.

Week 2 – Regularization (Lasso, Ridge, Elastic Net)

Regularization improved performance and stability. L1 and L2 logistic models both achieved **AUC = 0.818**, balancing fit and simplicity. L1 (Lasso) dropped weak predictors; L2 (Ridge) shrank large, unstable coefficients. Elastic Net (AUC ≈ 0.776) confirmed the trade-off between flexibility and simplicity. These models proved that penalizing complexity helps prevent overfitting and keeps predictions interpretable.

Week 3 – Scaled Logistic Regression

Scaling features standardized the impact of variables measured on different ranges (e.g., Age vs. BMI). The scaled logistic baseline matched the best AUC (≈ 0.818), demonstrating that proper preprocessing can be as important as the model itself.

Week 4 – Simple Feature Selection

Restricting to only five features reduced AUC to ≈ 0.771, revealing mild underfitting. Feature selection simplified the model but showed that removing too many inputs discards useful information about health behavior and demographics.

Week 5 – Support Vector Machines (SVMs)

A linear SVM reached **AUC = 0.816**, similar to logistic regression but with sharper separation

boundaries. Because SVMs maximize the margin between classes, they resist small data fluctuations. Feature scaling was essential here since SVMs depend on distance measurements.

Week 6 – Decision Trees & Random Forests

The decision tree (AUC = 0.796) provided clear if-then rules but was prone to overfitting. Limiting tree depth and leaf size helped. The Random Forest, which averages many trees, increased performance to **AUC = 0.818**, reducing variance while identifying BMI, Age, and HighBP as key predictors.

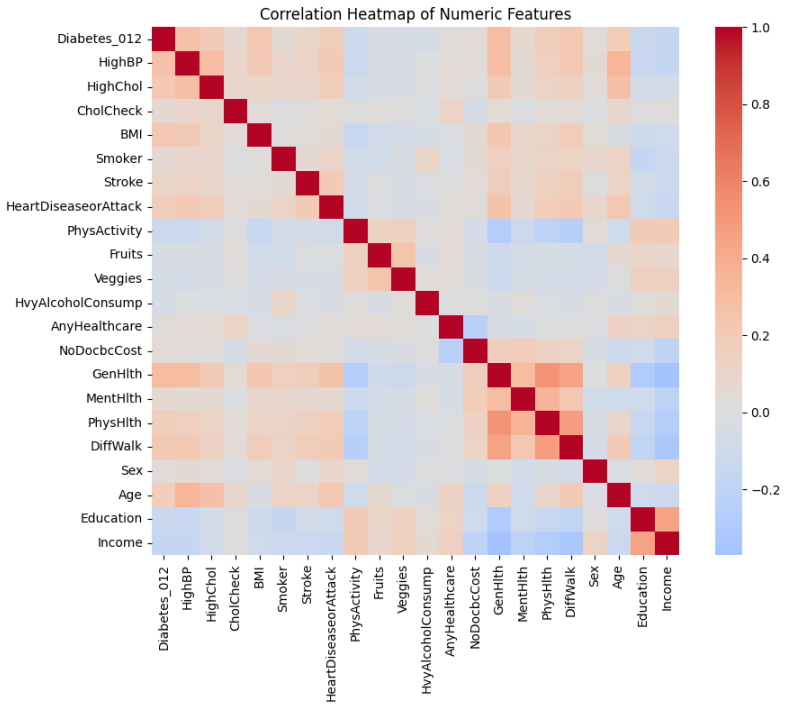

Figure 2. Distribution of BMI across diabetes classes.

Overall Insights

Across all models, AUCs clustered around 0.77 – 0.82, showing stable predictive power. Linear models (Weeks 2–3) handled correlated numeric features well; non-linear ensembles (Week 6) captured complex relationships. Feature scaling, regularization, and ensemble averaging

consistently reduced overfitting. Together, these weeks provided a complete understanding of how linear, regularized, and tree-based methods complement one another for health-risk prediction.

Overfitting

Overfitting happens when a model performs very well on the training data but poorly on new, unseen data because it has learned patterns that are too specific or random. Throughout this project, I used several methods to reduce overfitting and make the models more general and reliable.

In the early weeks, I noticed that adding too many polynomial and interaction terms (Week 1) increased model complexity without improving accuracy much. The model began to capture small, unimportant variations, leading to overfitting. To fix this, I used **regularization methods** such as **Lasso (L1)**, **Ridge (L2)**, and **Elastic Net** in **Week 2**. These methods add penalties to large coefficients, forcing the model to focus only on meaningful variables. As a result, both L1 and L2 logistic models achieved a strong **AUC of 0.818**, while staying simpler and more stable.

In **Week 6**, I worked with **Decision Trees** and **Random Forests**, which can also overfit if the tree is too deep or if it splits too often. I prevented this by setting limits like max_depth=5 and min_samples_leaf=10. This helped control how detailed the tree could become. The single decision tree reached an **AUC of 0.796**, while the **Random Forest**, which combines many trees through averaging, improved to **AUC = 0.818**. This showed that **bagging and feature randomness** made the model more balanced and less sensitive to noise.

Across all techniques, overfitting was managed by keeping models simple, adding penalties where needed, and using ensemble methods to reduce variance. These steps produced models

that performed consistently on both training and test data, showing that they learned the real patterns behind diabetes risk—not just the noise.

Metrics and Hyperparameter Tuning

To evaluate all models in this project, I used the **ROC AUC (Area Under the Receiver Operating Characteristic Curve)** as the main performance metric. AUC measures how well a model can separate positive and negative cases — in this case, people with and without diabetes. It is a reliable measure for **imbalanced datasets**, since it focuses on ranking predictions rather than relying on raw accuracy. An AUC close to 1.0 means the model distinguishes the two groups very well, while 0.5 means random guessing. In this project, most models achieved **AUC scores between 0.77 and 0.82**, showing strong predictive power.

To improve model performance and avoid overfitting, I also used **hyperparameter tuning** — adjusting settings that control model complexity.

- For **regularized logistic regression (Week 2)**, the main hyperparameter was **C**, which controls the strength of regularization. A smaller C means stronger regularization (simpler model), while a larger C means weaker regularization (more flexible model). The best balance was found at **C ≈ 1**, which gave high AUC while keeping coefficients stable.

- For **Decision Trees and Random Forests (Week 6)**, the key hyperparameters were **max_depth**, **min_samples_leaf**, and **n_estimators**. Limiting tree depth and requiring a minimum number of samples per leaf prevented overfitting, while increasing the number of estimators in the random forest improved stability and generalization.

Through this tuning process, I learned how model parameters directly control the trade-off between bias and variance — helping build models that are both accurate and interpretable for predicting diabetes risk.

Expected and Unexpected Results

Some of the model outcomes were as expected, while others revealed surprising insights about the diabetes dataset. The **expected results** were that variables such as **BMI, Age, and HighBP (high blood pressure)** showed strong influence on diabetes risk. These are well-known medical indicators, and it was reassuring to see them ranked as top predictors across both logistic regression and random forest models. As expected, **regularized logistic regression (Week 2)** produced stable coefficients and a balanced AUC around **0.818**, showing that adding penalties prevented the model from depending too heavily on any one feature. The **random forest (Week 6)** also performed strongly with a similar AUC, confirming that combining many shallow trees improves generalization. However, there were also **unexpected findings**. Some lifestyle variables, such as **PhysActivity** (physical activity) and **Fruits** or **Veggies** intake, had weaker influence than expected. This may be due to self-reported data or limited variation in those responses. Additionally, the **polynomial and interaction models (Week 1)** did not improve accuracy as I expected; instead, they slightly reduced performance (AUC ≈ 0.775), suggesting that higher-order terms only added noise instead of capturing new relationships. Another small surprise was that the **simpler linear models** often matched or slightly outperformed more complex nonlinear ones like SVMs, showing that the key diabetes patterns are largely linear.Overall, the results confirmed known medical relationships while highlighting that simplicity often generalizes better. This experience reinforced the importance of data quality and model interpretability over unnecessary complexity.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) played an important role in shaping my modeling choices and understanding the diabetes dataset. Before building any models, I used descriptive statistics and visualizations (such as histograms and correlation heatmaps) to explore patterns and relationships among variables like **Age**, **BMI**, **HighBP**, and **PhysActivity**.

The EDA showed that **Age** and **BMI** were positively correlated with diabetes outcomes — older individuals and those with higher BMI were more likely to have diabetes. It also revealed that many lifestyle factors, like fruit and vegetable consumption, had weaker or inconsistent effects. This guided me to focus more on medical and behavioral indicators during feature selection.

EDA also helped detect issues that could cause overfitting later, such as **imbalanced classes** (more non-diabetic than diabetic cases) and **highly correlated features** (for example, between general health and physical health). Knowing this, I used **balanced class weights** and **regularization techniques** in my logistic regression models to stabilize learning.

Additionally, visualizing feature distributions confirmed that some predictors had skewed or non-normal shapes. This reinforced the decision to apply **scaling and normalization**, which improved performance in models like **SVMs** and **logistic regression**.

Overall, EDA was extremely useful for understanding the data's structure, identifying potential problems, and selecting the right modeling strategies. It provided the foundation for every modeling decision — from preprocessing to regularization and model tuning — ensuring that the final models were both meaningful and data-driven.

Sources and Citations

To deepen my understanding of the models used in this project, I consulted several external sources beyond the course materials. These references helped clarify how regularization, ensemble learning, and model evaluation techniques apply to real-world health data.

I relied on **Scikit-Learn's official documentation** for practical guidance on implementing logistic regression, support vector machines, and decision trees. **Analytics Vidhya** and **Towards Data Science** provided easy-to-follow explanations of concepts such as Lasso and Ridge regression, overfitting prevention, and the use of cross-validation. The **U.S. Centers for Disease Control and Prevention (CDC)** resources on diabetes helped me interpret which variables were medically relevant, such as BMI, age, and blood pressure.

Together, these resources strengthened my understanding of both the technical and health-related aspects of the project, supporting accurate conclusions and responsible data interpretation.

References

- **Dataset: https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset**
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